No stage of the cumulative hierarchy contains itself: V_alpha is not in V_alpha

The Open Logic Project contributors

openlogic:set-theory/spine:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-spine/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: all three ordinal cases appear with their unfolding lines; the author's parenthetical '(twice)' and 'as before' style annotations are kept verbatim.

Editorial note. The label Valphanotref names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Transitivity, potency and cumulation of the V-hierarchy. Same unit, \ollabel{Valphabasicprops} (source line 847). Line 892: V_alpha in its successor stage; lines 899-901: at limits, V_alpha in V_beta together with V_beta in V_alpha forces V_beta in V_alpha's own membership chain.
- Successor and limit hierarchy equations. Same unit, the definition of V_alpha by transfinite recursion: V_succ = powerset, V_limit = union. Each case opens by unfolding the corresponding equation.

Source reference. Original reference Valphabasicprops: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 847–847. Prerequisite pointer invoked at proof line(s) 893; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *For each ordinal α, $V_\alpha \notin V_\alpha$.*

Proof. By transfinite induction. Evidently $V_\emptyset \notin V_\emptyset$.

If $V_{+1\alpha} \in V_{+1\alpha} = \mathcal{P}V_\alpha$, then $V_{+1\alpha} \subseteq V_\alpha$; and since $V_\alpha \in V_{+1\alpha}$ by Valphabasicprops, we have $V_\alpha \in V_\alpha$. Conversely: if $V_\alpha \notin V_\alpha$ then $V_{+1\alpha} \notin V_{+1\alpha}$

If α is a limit and $V_\alpha \in V_\alpha = \bigcup_{\beta\in\alpha} V_\beta$, then $V_\alpha \in V_\beta$ for some $\beta \in \alpha$; but then also $V_\beta \in V_\alpha$ so that $V_\beta \in V_\beta$ by Valphabasicprops (twice). Conversely, if $V_\beta \notin V_\beta$ for all $\beta \in \alpha$, then $V_\alpha \notin V_\alpha$. □